

PHYS 202 Discussion Section Worksheet

Week 2: Wave Properties — Complete Solutions

Algebra-based solutions with all numerical work shown

The following relations are used throughout the worksheet:

$$\mu = \frac{m}{L}, \quad v = \sqrt{\frac{F_T}{\mu}} = \sqrt{\frac{F_T L}{m}}, \quad f_n = n f_1, \quad f_1 = \frac{1}{2L} \sqrt{\frac{F_T}{\mu}}.$$

Here μ is linear mass density, F_T is string tension, v is wave speed, and f_n is the frequency of the n th harmonic.

Section A: Wave Types and Properties

1. Transverse wave

Define a transverse wave in your own words and give an example.

Answer. A transverse wave is a wave for which the medium oscillates perpendicular to the direction in which the wave travels. For example, if a wave pulse travels horizontally along a stretched string, each small piece of the string moves up and down. Light is another example of a transverse wave.

The important comparison is between two directions:

$$\text{direction of oscillation} \perp \text{direction of wave travel}.$$

2. Longitudinal wave

Define a longitudinal wave in your own words and give an example.

Answer. A longitudinal wave is a wave for which the medium oscillates parallel to the direction in which the wave travels. For example, as a sound wave travels through air, the air molecules move back and forth along the direction of travel. This motion produces alternating compressions and rarefactions.

In this case,

$$\text{direction of oscillation} \parallel \text{direction of wave travel}.$$

3. Changes to the speed of a wave on a string

A wave pulse travels along a string at an initial speed $v = 10 \text{ m/s}$. Determine the new wave speed for each independent change.

For a wave on a string,

$$v = \sqrt{\frac{F_T}{\mu}}, \quad \mu = \frac{m}{L}.$$

Combining the equations gives

$$v = \sqrt{\frac{F_T}{m/L}} = \sqrt{\frac{F_T L}{m}}.$$

This form makes the effects of changing tension, mass, or length explicit.

(a) The string's tension is doubled

The new tension is $F'_T = 2F_T$, while m and L remain unchanged. Therefore,

$$\begin{aligned} v' &= \sqrt{\frac{F'_T L}{m}} \\ &= \sqrt{\frac{(2F_T) L}{m}} \\ &= \sqrt{2} \sqrt{\frac{F_T L}{m}} \\ &= \sqrt{2} v. \end{aligned}$$

Using $v = 10 \text{ m/s}$,

$$\begin{aligned} v' &= \sqrt{2} (10 \text{ m/s}) \\ &= 14.142 \dots \text{ m/s} \\ &\approx 14.1 \text{ m/s}. \end{aligned}$$

$v' \approx 14.1 \text{ m/s}$

(b) The string's mass is quadrupled

The new mass is $m' = 4m$, while F_T and L remain unchanged. The new linear mass density is

$$\mu' = \frac{m'}{L} = \frac{4m}{L} = 4\mu.$$

Thus,

$$\begin{aligned} v' &= \sqrt{\frac{F_T}{\mu'}} \\ &= \sqrt{\frac{F_T}{4\mu}} \\ &= \frac{1}{2} \sqrt{\frac{F_T}{\mu}} \\ &= \frac{v}{2}. \end{aligned}$$

Numerically,

$$v' = \frac{10 \text{ m/s}}{2} = 5.0 \text{ m/s}.$$

$$v' = 5.0 \text{ m/s}$$

(c) The string's length is quadrupled

The new length is $L' = 4L$, while F_T and the total string mass m remain unchanged. The new linear mass density is

$$\mu' = \frac{m}{L'} = \frac{m}{4L} = \frac{\mu}{4}.$$

Therefore,

$$\begin{aligned} v' &= \sqrt{\frac{F_T}{\mu'}} \\ &= \sqrt{\frac{F_T}{\mu/4}} \\ &= \sqrt{4} \sqrt{\frac{F_T}{\mu}} \\ &= 2v. \end{aligned}$$

Numerically,

$$v' = 2(10 \text{ m/s}) = 20 \text{ m/s}.$$

$$v' = 20 \text{ m/s}$$

(d) The string's mass and length are both quadrupled

Now $m' = 4m$ and $L' = 4L$. The new linear mass density is

$$\begin{aligned} \mu' &= \frac{m'}{L'} \\ &= \frac{4m}{4L} \\ &= \frac{m}{L} \\ &= \mu. \end{aligned}$$

Because both the tension and the linear mass density are unchanged,

$$\begin{aligned} v' &= \sqrt{\frac{F_T}{\mu'}} \\ &= \sqrt{\frac{F_T}{\mu}} \\ &= v. \end{aligned}$$

Hence,

$$v' = 10 \text{ m/s}$$

4. LIGO and interference

What should scientists look for in the signal to determine whether a gravitational wave is stretching and shrinking the detector arms?

The laser is split into two beams. The beams travel down perpendicular arms, reflect from mirrors, return, and recombine. When the two returning waves are combined, their electric-field disturbances obey the principle of superposition.

If both arms have the expected relative lengths, the waves return with a known relative alignment. Depending on that alignment, they may interfere constructively, destructively, or somewhere between those two limits.

A passing gravitational wave stretches one arm while shrinking the other. The two beams then travel slightly different distances, so their crests and troughs no longer arrive with exactly the same relative alignment. The interference at the photodetector changes, which changes the detected light intensity.

Answer. Scientists should look for a small, time-dependent change in the interference signal—equivalently, a change in the light intensity measured at the photodetector. The signal is caused by a changing difference between the two optical path lengths, not by the formation of a standing wave.

Section B: Waves Problem Solving

1. Helicopter and hiker

The given quantities are

$m_h = 80.0 \text{ kg}$	(hiker mass),
$m_s = 70.0 \text{ kg}$	(sling mass),
$m_c = 8.00 \text{ kg}$	(cable mass),
$L_c = 15.0 \text{ m}$	(cable length),
$\Delta t = 0.250 \text{ s}$	(pulse travel time).

(a) Determine the wave speed v

The pulse travels one cable length, so its travel distance is $\Delta x = L_c = 15.0 \text{ m}$. Therefore,

$$\begin{aligned} v &= \frac{\Delta x}{\Delta t} \\ &= \frac{15.0 \text{ m}}{0.250 \text{ s}} \\ &= 60.0 \text{ m/s.} \end{aligned}$$

$$v = 60.0 \text{ m/s}$$

(b) Determine the cable tension F_T

First find the cable's linear mass density. Only the cable's mass and length belong in μ :

$$\begin{aligned}\mu &= \frac{m_c}{L_c} \\ &= \frac{8.00 \text{ kg}}{15.0 \text{ m}} \\ &= 0.533333 \dots \text{ kg/m}.\end{aligned}$$

The wave speed on the cable is

$$v = \sqrt{\frac{F_T}{\mu}}.$$

Square both sides and solve for the tension:

$$\begin{aligned}v^2 &= \frac{F_T}{\mu}, \\ \mu v^2 &= F_T, \\ F_T &= \mu v^2.\end{aligned}$$

Substitute the known values:

$$\begin{aligned}F_T &= (0.533333 \dots \text{ kg/m}) (60.0 \text{ m/s})^2 \\ &= (0.533333 \dots \text{ kg/m}) (3600 \text{ m}^2/\text{s}^2) \\ &= 1920 \text{ kg m/s}^2 \\ &= 1.92 \times 10^3 \text{ N}.\end{aligned}$$

$$F_T = 1.92 \times 10^3 \text{ N}$$

(c) Write Newton's second law

The cable tension acts directly on the sling, while the sling supports the hiker. The cleanest system is therefore the *hiker and sling together*. The total mass of this system is

$$\begin{aligned}M &= m_h + m_s \\ &= 80.0 \text{ kg} + 70.0 \text{ kg} \\ &= 150.0 \text{ kg}.\end{aligned}$$

Choose upward as the positive direction. The external vertical forces on the combined system are the upward cable tension F_T and the downward weight Mg . Newton's second law gives

$$\begin{aligned}\sum F_y &= Ma, \\ F_T - Mg &= Ma.\end{aligned}$$

$$F_T - Mg = Ma$$

If one chose the hiker alone as the system, the upward force would be the contact force from the sling rather than the cable tension. Combining the hiker and sling removes that unknown internal force.

(d) Determine the acceleration

Starting from

$$F_T - Mg = Ma,$$

solve algebraically for a :

$$\begin{aligned} F_T - Mg &= Ma, \\ \frac{F_T - Mg}{M} &= a, \\ a &= \frac{F_T}{M} - g. \end{aligned}$$

Using $g = 9.80 \text{ m/s}^2$,

$$\begin{aligned} a &= \frac{1.92 \times 10^3 \text{ N}}{150.0 \text{ kg}} - 9.80 \text{ m/s}^2 \\ &= 12.8 \text{ m/s}^2 - 9.80 \text{ m/s}^2 \\ &= 3.00 \text{ m/s}^2. \end{aligned}$$

The positive sign means that the acceleration is upward.

$$a = 3.00 \text{ m/s}^2 \text{ upward}$$

2. Piano strings

The fundamental frequencies are

$$f_C = 262 \text{ Hz}, \quad f_A = 440 \text{ Hz}.$$

(a) Find the next two harmonics of the C string

For an ideal string fixed at both ends, the harmonic frequencies are whole-number multiples of the fundamental:

$$f_n = nf_1.$$

The next two harmonics after the fundamental are the second and third harmonics. Thus,

$$\begin{aligned} f_2 &= 2f_C = 2(262 \text{ Hz}) = 524 \text{ Hz}, \\ f_3 &= 3f_C = 3(262 \text{ Hz}) = 786 \text{ Hz}. \end{aligned}$$

$$f_2 = 524 \text{ Hz}, \quad f_3 = 786 \text{ Hz}$$

(b) Determine the ratio of the string tensions

The fundamental frequency of a string is

$$f_1 = \frac{1}{2L} \sqrt{\frac{F_T}{\mu}}.$$

Apply this equation separately to the A and C strings:

$$f_A = \frac{1}{2L_A} \sqrt{\frac{F_{T,A}}{\mu_A}},$$

$$f_C = \frac{1}{2L_C} \sqrt{\frac{F_{T,C}}{\mu_C}}.$$

The problem states that the two strings have the same length and the same linear mass density, so

$$L_A = L_C = L, \quad \mu_A = \mu_C = \mu.$$

Take the ratio of the two frequency equations:

$$\begin{aligned} \frac{f_A}{f_C} &= \frac{\frac{1}{2L} \sqrt{F_{T,A}/\mu}}{\frac{1}{2L} \sqrt{F_{T,C}/\mu}} \\ &= \sqrt{\frac{F_{T,A}}{F_{T,C}}}. \end{aligned}$$

Square both sides:

$$\left(\frac{f_A}{f_C}\right)^2 = \frac{F_{T,A}}{F_{T,C}}.$$

Substitute the given frequencies:

$$\begin{aligned} \frac{F_{T,A}}{F_{T,C}} &= \left(\frac{440 \text{ Hz}}{262 \text{ Hz}}\right)^2 \\ &= (1.679389 \dots)^2 \\ &= 2.82035 \dots \\ &\approx 2.82. \end{aligned}$$

$$\frac{F_{T,A}}{F_{T,C}} = 2.82$$

Therefore, under the assumptions in the problem, the A string has about 2.82 times the tension of the C string. Equivalently,

$$\frac{F_{T,C}}{F_{T,A}} = \frac{1}{2.82} \approx 0.355.$$